



Lab 12

MOSFET Current Source and Current Mirrors

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12.1 Objective

To design MOSFET current source and current mirror circuits and utilize current-mirror circuit for biasing of MOSFET CS amplifier.

Task 1: Pre-Lab

Q1. Why do we use MOSFETs as a Current Source, Explain?

MOSFETs are used as current sources because they can provide a stable and predictable current regardless of variations in voltage or load resistance. This is achieved by operating the MOSFET in the saturation region, where the drain current becomes independent of the drain-source voltage and depends only on the gate-source voltage.

Q2. What is MOSFET Current Mirror? What is the significance of using Current Mirrors?

A MOSFET Current Mirror is a circuit designed to copy (mirror) a reference current from one branch to another while maintaining the same current value. It consists of two or more MOSFETs, where one MOSFET is used to set the reference current, and the others replicate this current. Significance of Using Current Mirrors: Consistent Biasing: Current mirrors ensure that multiple transistors in a circuit receive the same bias current, improving performance. Power Efficiency: Instead of using multiple resistors for biasing, a current mirror efficiently distributes current to different circuit parts. Improved Performance in Analog Circuits: Used in differential amplifiers, op-amps, and active loads, current mirrors enhance gain and reduce variations due to power supply changes. Integrated Circuit Applications: Commonly used in ICs to provide stable current sources with minimal external components.

12.2 Introduction to Channel Length Modulation

From the ideal characteristics curves of MOSFET, in saturation region drain current is independent of V_{DS} . Thus, a change in V_{DS} a zero change in I_D , which implies that the incremental resistance looking into the drain of a saturated MOSFET is infinite. This, however, is an idealization based on the premise that once the channel is pinched off at the drain end, further increases in V_{DS} have no effect on the channel's shape. But, in practice, increasing V_{DS} beyond V_{OV} does affect the channel somewhat. Specifically, as V_{DS} is increased, the channel pinch-off point is moved slightly away from the drain, toward the source. This is illustrated in Figure 1, from which we note that the voltage across the channel remains constant at V_{OV} , and the additional voltage applied to the drain appears as a voltage drop across the narrow depletion region between the end of the channel and the drain region.

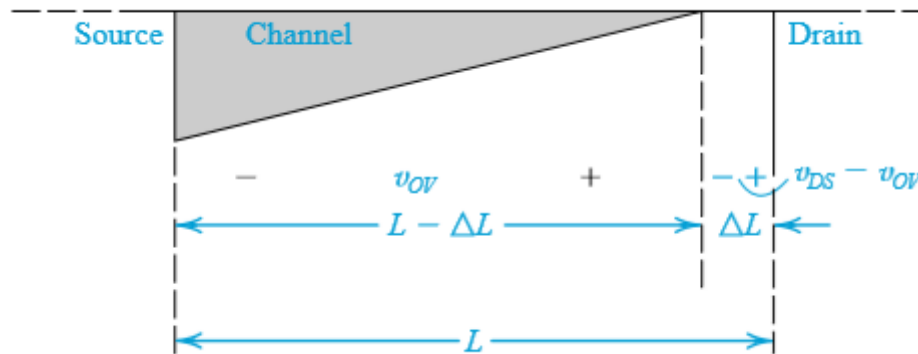


Figure 1: Channel-length modulation

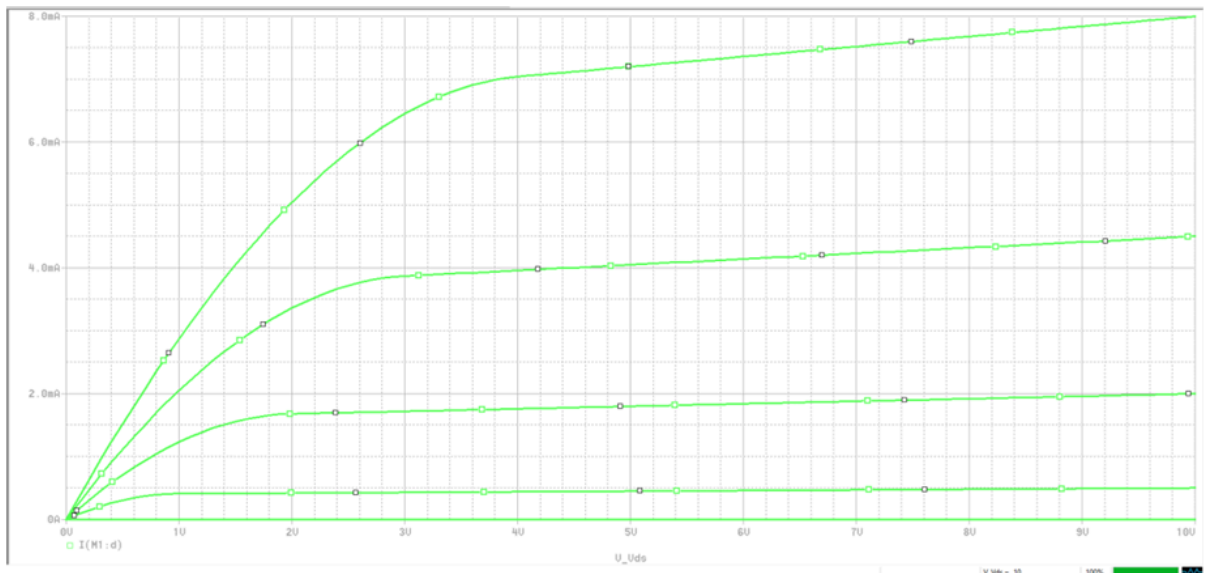
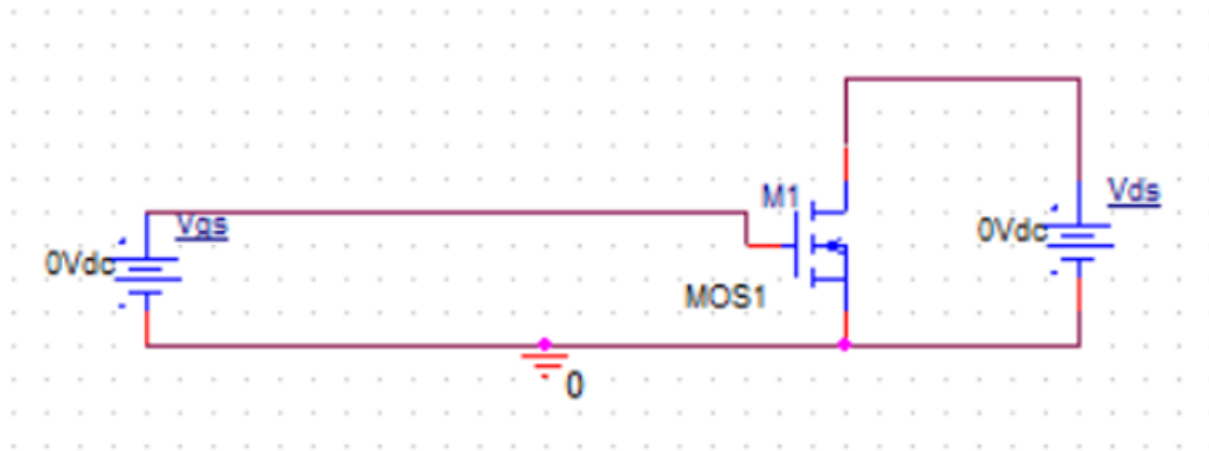
This voltage accelerates the electrons that reach the drain end of the channel and sweeps them across the depletion region into the drain. Note, however, that (with depletion-layer widening) the channel length is in effect reduced, from L to $L - \Delta L$, a phenomenon known as **channel-length modulation**. Now, since I_D is inversely proportional to the channel length, I_D increases with V_{DS} . This effect is accounted in drain current equation in saturation region by including channel-length modulation coefficient λ .

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 (1 + \lambda V_{DS})$$

Task 2: To incorporate channel length modulation in MOSFET PSpice model and obtain output characteristic curves

The channel-length modulation coefficient λ (V^{-1}) is modelled as PSpice parameter **LAMBDA** in model editor like other parameters **VTO** (Threshold Voltage), **KP** (Process trans-conductance), **W** (width) and **L** (length).

1. Add an MbreakN3 part from PSpice library or use discrete NMOS model. Edit the model using PSpice Model Editor to specify the following given specifications:
 - Threshold voltage: 1V
 - Process trans-conductance: 0.8m A/V²
 - Length: 0.8 μ m
 - Width: 0.8 μ m
 - Channel-length modulation coefficient: 0.025 V⁻¹
2. Obtain characteristic curves (I_D - V_{DS}) for the modelled MOSFET by sweeping drain to source voltage from 0 to 10V for the different gate-to-source voltages (1 to 5V with step-size of 1V).
3. Add captures of the obtained plot.



4. For each of the obtained curves, find the value of output resistance r_o using the formula given below. Complete Table 1.

$$r_o = \left. \frac{\partial V_{DS}}{\partial I_D} \right|_{V_{GS}=\text{constant}} = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS}=\text{constant}}$$

Table 1: Effect of channel length modulation on output resistance of MOSFET

Gate-to-Source Voltage V_{GS}	Output Resistance: r_o
2V	100000
3 V	25275
4 V	11233.3
5 V	6240.7

Reon
Energy $\Rightarrow$ For $V_{GS} = 2V$:

$$r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{4.1011 - 4}{441.1011\mu - 440\mu} = \boxed{100\text{ k}\Omega}$$

 $\Rightarrow$ For $V_{GS} = 3V$:

$$r_o = \frac{4.1011 - 4}{1.769\text{m} - 1.76\text{m}} = \boxed{25275\Omega}$$

 $\Rightarrow$ For $V_{GS} = 4V$:

$$r_o = \frac{4.1011 - 4}{3.969\text{m} - 3.96\text{m}} = \boxed{11233.3\Omega}$$

 $\Rightarrow$ For $V_{GS} = 5V$:

$$r_o = \frac{4.1011 - 4}{7.0562\text{m} - 7.04\text{m}} = \boxed{6240.7\Omega}$$

5. How are these curves different from the ones obtained in previous labs? Is there any effect of channel-length modulation on MOS current in saturation region? What is the effect of gate-to-source voltage on output resistance?

Unlike ideal MOSFET curves where the saturation region appears as perfectly horizontal lines (constant current regardless of V_{DS}), these curves show a positive slope in the saturation region due to channel-length modulation. Channel-length modulation ($\lambda = 0.025$) causes the drain current to increase with increasing V_{DS} in the saturation region. This is because as V_{DS} increases, the effective channel length decreases slightly, which reduces the channel resistance and allows more

current to flow. The effect of gate-to-source voltage on output resistance is inverse, as V_{GS} increases, output resistance decreases dramatically.

12.3 MOSFET as Current Source

An ideal current source is a two-terminal circuit which provides a constant and stable output current value regardless of the voltage across the output load. The current supplied by such a source is independent of the voltage across source terminals. Real current sources, however, present a slope in their voltage-current characteristics. The value of this slope is given by the conductivity of the **source resistance** placed in parallel with the source. The source resistance is not physically present in the device but it is rather a way to represent the non-ideal output of current source in model. This is shown in Figure 2. The ideal current source has an infinite resistance whereas, for a practical current source, the resistance is not infinite but large in magnitude.

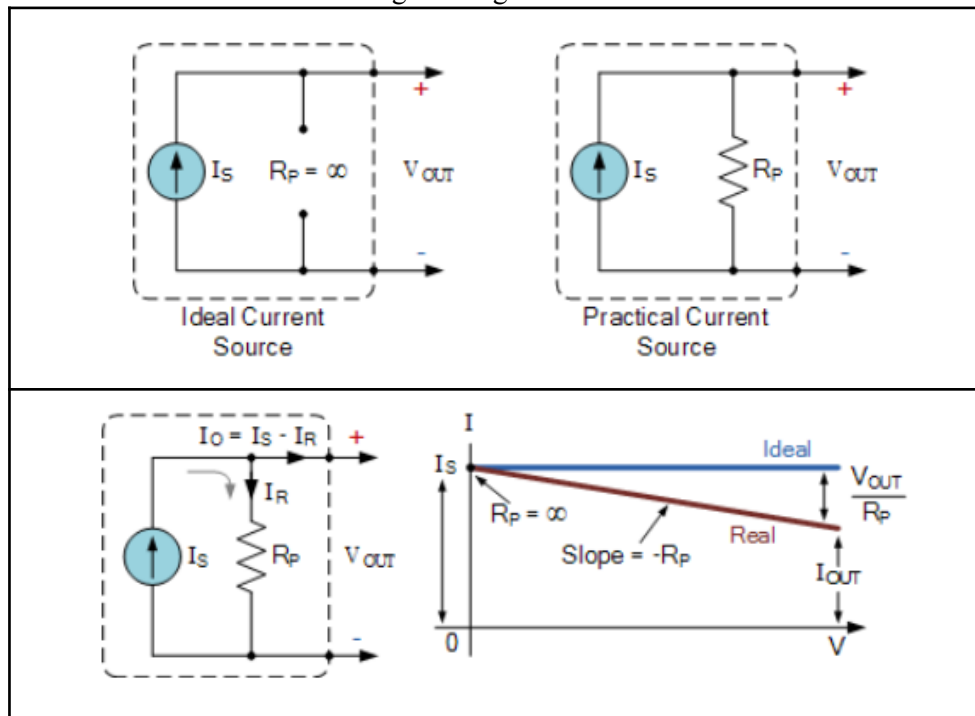


Figure 2: Ideal and practical current sources

From the drain characteristics of MOSFET, we can conclude that a MOSFET can be used as a current source when operated in saturation region. The drain current depends on gate-to-source voltage. Therefore, MOSFET in saturation region acts as voltage-controlled current source.

In the graph of Figure 2, as the load voltage increases, the supply-current of a practical source decreases. When MOSFET is used as a current-source, the load is connected at drain terminal. If load voltage increases due to loading of source, the drain-to-source voltage of MOSFET will decrease. Comparing with the drain-characteristic curve of MOSFET, as drain-to-source voltage decreases, drain current will decrease if channel-length modulation is taken into account i.e. the MOS acts as practical current source. Else, the drain current will remain constant provided the transistor keeps operating in saturation region i.e. the transistor behaves as ideal current source.

The circuit shown in Figure 3 has a diode-connected MOSFET. In previous labs, we have seen that this MOSFET operates in saturation region as its gate and drain terminals are tied together. To design this circuit for any required value of drain current, the required gate-to-source voltage is first calculated using,

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$$

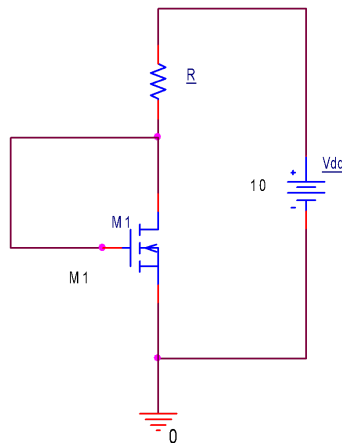


Figure 3: Diode connected MOSFET to establish reference current for current source

As gate-to-source voltage is equal to drain-to-source voltage here, therefore, this is the required drain voltage. The required drain current can be established by using a particular value of resistance R which is found by using $V_{DS}=V_{GS}$ in the given KVL equation for drain to source loop.

$$V_D = V_{DD} - I_D R$$

Task 3: To establish reference current for MOSFET current source

1. Edit the modelled MOSFET in Task 1 to ignore the effect of channel length modulation.
2. When this MOSFET is to be used in circuit of Figure 3, calculate the required gate-to-source voltage and resistance to obtain 1mA drain current. Let, V_{DD} be 10V.



Reon
Energy

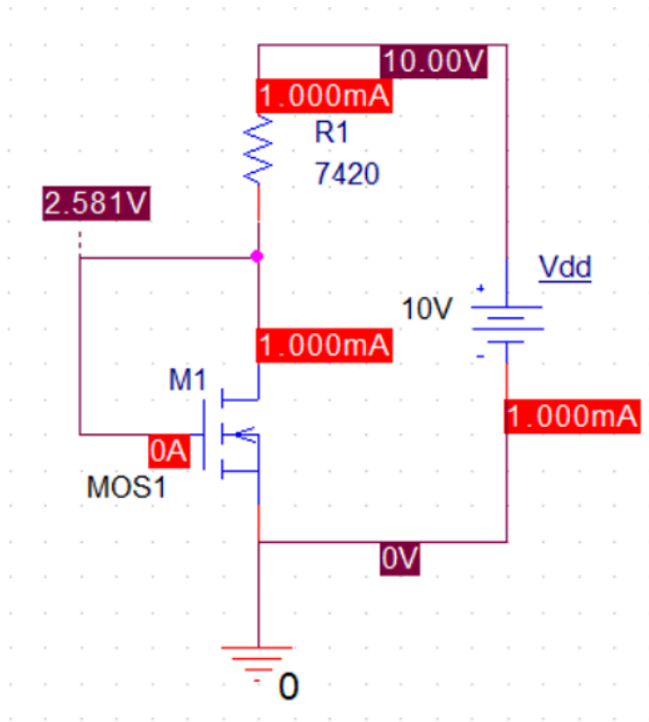
Task 3 :

$$\rightarrow I_D = \frac{1}{2} \frac{k_n W}{L} (V_{GS} - V_t)^2$$
$$\rightarrow \frac{2 I_D L}{k_n W} = (V_{GS} - V_t)^2$$
$$\rightarrow \sqrt{\frac{2 I_D L}{k_n W}} + V_t = V_{GS}$$
$$\Rightarrow V_{GS} = \sqrt{\frac{2(1\text{m})(0.8\mu)}{(0.8\mu)(0.8\mu)}} + 1 = 2.5811\text{V}$$
$$\Rightarrow V_{DS} = V_{DD} - I_D R$$
$$R = \frac{V_{DD} - V_{DS}}{I_D} = \frac{10 - 2.5811}{1 \times 10^{-3}}$$
$$R = 7418.9 \Omega$$

Table 2: Data-collection table for circuit shown in Figure 3

Parameters	Calculated Values	Simulated Results
Drain Current: I_D	1mA	1mA
Resistance: R	7418.9	7420
Gate-to-Source Voltage: V_{GS}	2.5811V	2.581V
Gate-to-Source Voltage: V_{DS}	2.5811V	2.581V

3. Using the obtained values, simulate the circuit and take measurements for Table 1.



12.4 Current Mirror Circuits with MOSFETs

The circuit shown in Figure 4 is called current-mirror circuit.

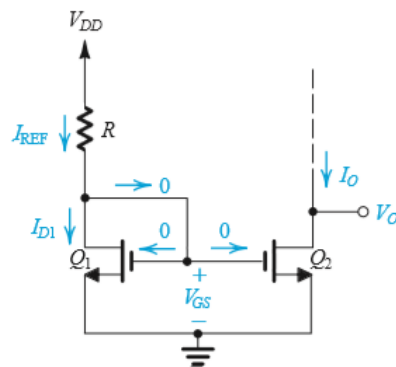


Figure 4: Current-mirror circuit

The heart of this circuit is Q1 transistor through which reference current is established as done in previous task.

$$I_{REF} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_1 (V_{GS} - V_t)^2$$

Another transistor Q2 is connected such that its gate and source terminals are connected with the gate and source terminals of Q1 transistor. If we assume that Q2 operates in **saturation** mode then, the current I_o will be

$$I_o = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_2 (V_{GS} - V_t)^2$$



If both the transistors have equal threshold voltage and process trans-conductance parameters, then the output current of Q2 can be related to the reference current by

$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

This is a simple and attractive relationship used to calculate current gain or current transfer ratio of this circuit. The special connection of Q1 and Q2 provides an output current I_O that is related to the reference current I_{REF} by the ratio of the aspect ratios of the transistors. In the special case of identical transistors $I_{REF} = I_O$, and the circuit simply replicates or mirrors the reference current in the output terminal. This has given this circuit the name of current mirror, a name that is used irrespective of the ratio of device dimensions.

I_O is the current that flows through any load connected at the drain of Q2. This load can be a resistor or any other circuit.

Task 4: To design MOSFET current mirror circuits

1. For the current mirror circuit shown in Figure 4, let Q1 be the MOS transistor modelled in previous task. The reference current has already been established i.e. 1mA.
2. Find the required aspect ratio for Q2 transistor that gives output drain current of 12mA.
3. Add another MOS model in PSpice with the parameters required to model Q2 transistor.
4. Modify the circuit implemented in Task 2 to add Q2¹. Q2 is to be connected with same supply through a load resistor of 200 Ohms.
5. Simulate the circuit and measure the output current. Measure this load current for different values of load resistor.²

¹ Do not copy paste the MOSFETs instead add a new MOSFET from component library (Place part).

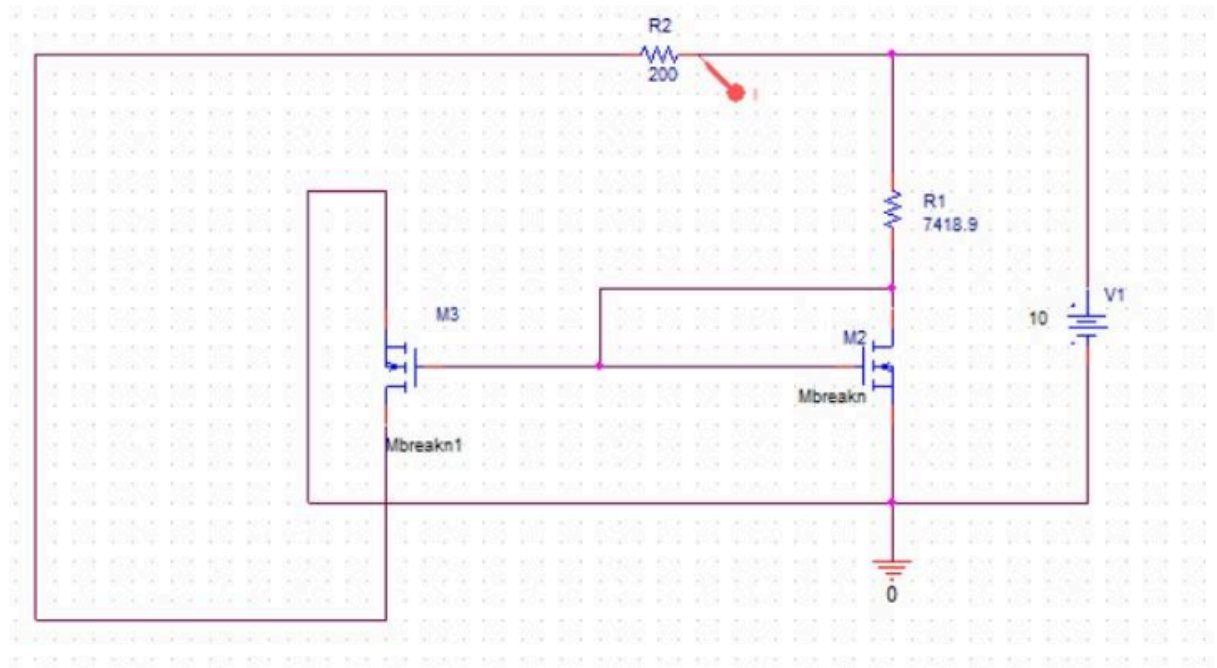


Table 3: Output current of current-mirror circuit at different loads

Output Current	Load Resistor (200 Ohms)	Load Resistor (500 Ohms)	Load Resistor (1k Ohms)
Drain Current: I_o	12mA	12mA	9.184mA

6. What is the maximum value of load resistor for which the transistor Q2 keeps operating in saturation region?

V_{DS} must be $\geq 2.58V - 1V = 1.58V$. Since $V_{DS} = V_{DD} - I \times R_L$; $R_L = (V_{DD} - 1.58V)/I = (10V - 1.58V)/12mA = 701.667\Omega$

7. Does this circuit provide the required constant current of 12mA to the load? Refer the results collected in last Table and use the maximum value of load resistor to interpret the required condition for correct working of MOS current-mirror circuit?

At 200 Ω : 12mA (higher than expected)

At 500 Ω : 12mA (close to expected)

At 1k Ω : 9.184mA (significantly lower)

For the current mirror to function properly, both transistors must remain in the saturation region. To ensure this, the load resistance should be kept below a critical threshold (approximately 702 Ω). If the load resistance exceeds this value, transistor enters the linear region, causing the current mirroring to fail.

12.5 Biasing MOSFET CS Amplifier by Current Source

In previous labs, we have been using voltage-divider circuit for biasing of CS amplifier. In this lab, we will bias CS amplifier circuit of Lab 10 using a current source. Remember, biasing is needed to establish certain operating point (I_D and V_D) for MOSFET and for its application as an amplifier; it needs to be biased in saturation region. Using constant current source is the most efficient method by which a MOSFET can be biased. A general representation for biasing of MOSFET using current source is shown in Figure 5.

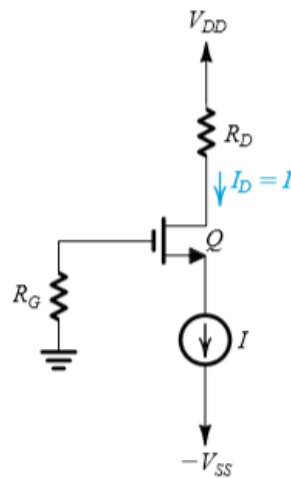


Figure 5: Biasing by constant current source

The current-mirror circuit implemented in Task 4 using MOSFETs can be used as current source for MOSFET biasing. Here, R_G resistor (usually in range of $M\Omega$) establishes a DC ground at the gate and presents a large input resistance to the input signal source which can be capacitively coupled to the gate. The complete CS amplifier circuit with coupling capacitors and by-pass is shown in Figure 6. The current source connected here for biasing the transistor is shown in Figure 7. The current mirror utilizes positive and negative sources (V_{DD} and $-V_{SS}$).

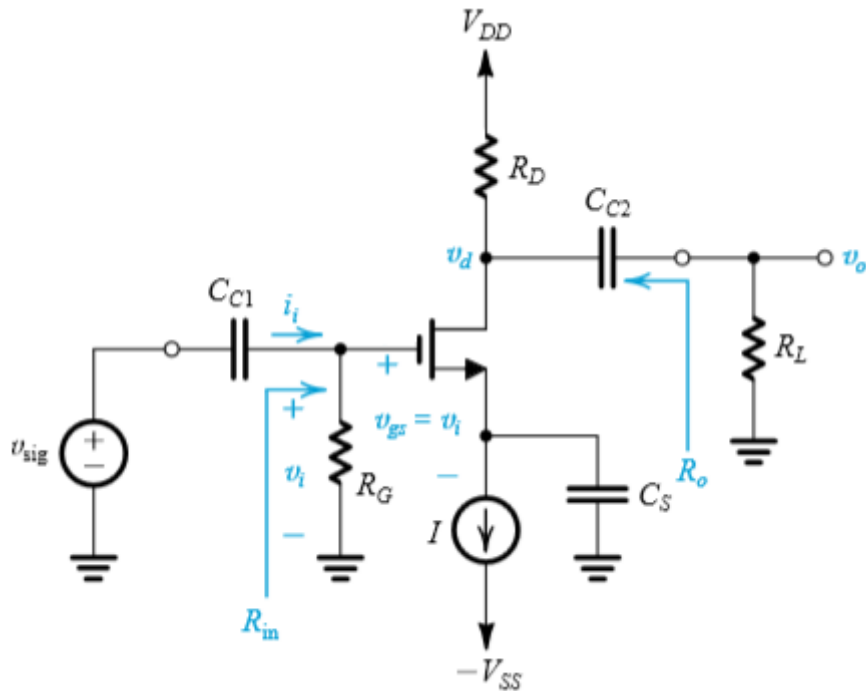


Figure 6: Common-source amplifier circuit with constant current source biasing

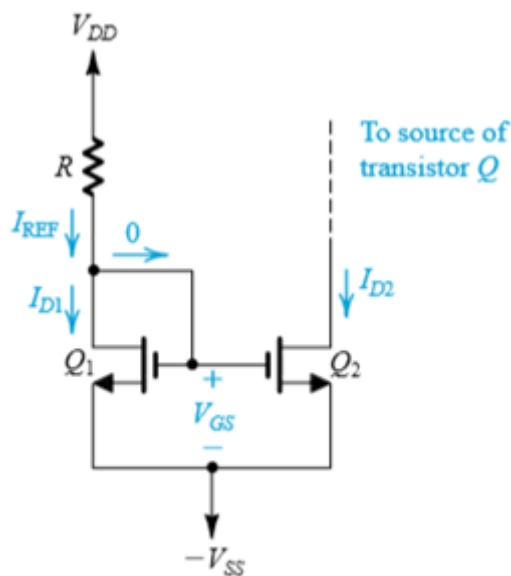


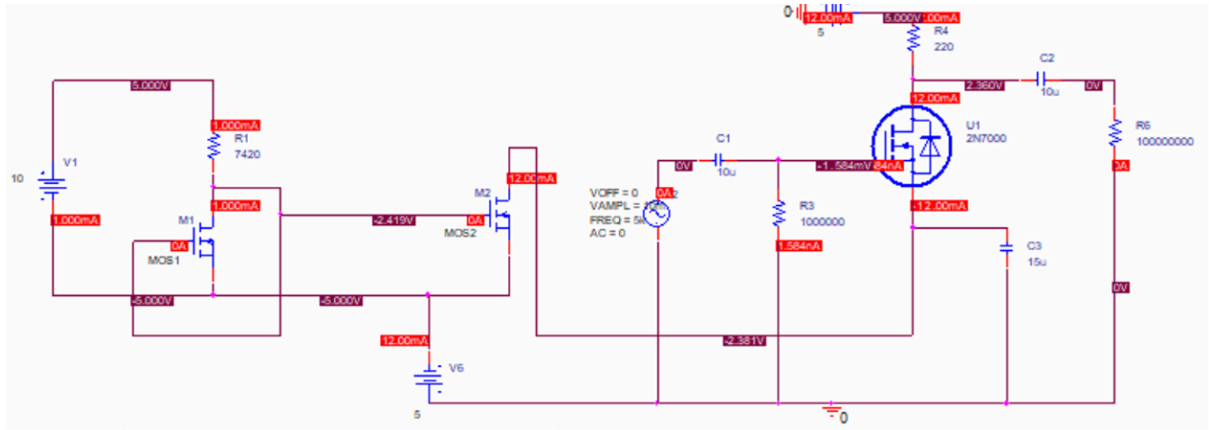
Figure 7: Constant-current source circuit to be used for CS amplifier biasing

Task 5: To design and implement CS amplifier circuit biased by current source

1. Implement the CS amplifier circuit shown in Figure 6 using 2N7000 MOSFET. Let, $R_G=1\text{M}\Omega$, $R_D=220\Omega$, $V_{DD}=5\text{V}$, $V_{SS}=-5\text{V}$, $C_1=C_2=10\mu\text{F}$ and $C_S=15\mu\text{F}$.
2. In place of current-source, connect the output terminal (drain terminal) of a current-mirror circuit (Figure 7) that establishes drain current of 12mA for the MOS amplifier.



3. Apply AC signal of 10mV (amplitude) at 5 kHz frequency at the input of this amplifier and measure unloaded output voltage. (To simulate open circuit, connect 100 Mega Ω in place of load resistor R_L).



4. Complete table 3 to characterize the implemented CS amplifier

Table 3: CS Amplifier biased by current source

Parameters	Values
Drain Current: I_D	12.09mA
Drain Voltage: V_D	2.340V
Source Voltage: V_S	-2.382V
Drain-to-Source Voltage: V_{DS}	4.722V



Gate-to-Source Voltage: V_{GS}	2.38V
Trans-conductance: $g_m = \frac{2I_D}{V_{GS} - V_T}$	0.103
Calculated Gain: $A_v = -g_m R_D$	-22.60
Measured Gain: A_v	20.95
Mode of Operation of Q (2N7000)	Saturation

Task 6: Post Lab

Q1. How can we improve the performance of a current mirror circuit?

The performance of a current mirror can be improved by using well-matched transistors, which ensures accurate current copying. Increasing the output resistance by using configurations like the cascode current mirror helps maintain a constant output current despite voltage changes. Adding emitter or source degeneration resistors improves thermal stability and matching. Using Wilson or other enhanced current mirrors introduces feedback that reduces error and enhances accuracy. Temperature compensation techniques and replacing resistive loads with active loads also help in achieving better performance.

Q2. What is Wilson Current Mirror? Which type of current mirror is used most?

A Wilson current mirror is an improved current mirror design that uses three transistors instead of two. It introduces feedback to force the output current to closely match the input current, resulting in better accuracy and a much higher output resistance. This configuration minimizes error and improves stability under varying conditions. Among the different types, the Wilson and cascode current mirrors are commonly used in integrated circuits due to their high performance, while the basic current mirror is still popular in simple applications because of its simplicity.



1.1 Assessment Rubric

Lab 12 MOSFET Current Source and Current Mirrors

Name:	Student ID:
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Points Distribution

Task No.	LR2 Simulation 32	LR 4 Data Collection 20	LR 5 Plots 8	LR 6 Calculation 8	LR 10 Analysis 24
Task 1					/8
Task 2	/8	/6	/4	-	/4
Task 3	/8	/4	-	/4	-
Task 4	/8	/4	-	/4	/8
Task 5	/8	/6	/4	-	/4
Task 6					/4
	/32	/20	/8	/8	/28
CLO Mapped	CLO 2	CLO 1	CLO 1	CLO 2	CLO 1

Affective Domain Rubric		Points	CLO Mapped
AR 4	Report Submission	/4	CLO 4

CLO	Total Points	Points Obtained
1	56	
2	40	
4	04	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.